

MD TASADUL ISLAM HEMAL

MSc Innovative Textile Development | Sustainability, Circularity & LCA in Textiles | Digital Product Passport (DPP)

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PROFESSIONAL SUMMARY

Textile engineer and MSc Innovative Textile Development candidate at Saxion University of Applied Sciences, with 6+ years of garment-industry supply chain and merchandising experience and active research in sustainability, circularity, Life Cycle Assessment (LCA), and Digital Product Passports (DPP) for the fashion value chain. Recently co-delivered a Digital Product Passport project for circular denim in collaboration with MUD Jeans (Netherlands) and Tejidos Royo (Valencia, Spain) — covering LCA inputs, PEFCR-aligned yarn mapping, supplier data collection, and stakeholder reporting. Combines hands-on supplier-side experience from one of the world's largest apparel sourcing countries with EU regulatory awareness (CSDDD, ESPR/DPP) and applied data tools (Python, GS1/EPCIS). Based in Enschede, NL, with a valid BSN and currently enrolled full-time at Saxion; available full-time (40 hrs/week) for a 6-month internship starting Aug/Sep 2026. Seeking sustainability roles in the Netherlands focused on supply chain mapping, supplier performance monitoring, LCA, and circular fashion.

EDUCATION

MSc Innovative Textile Development (EQF Level 7)

Sep 2025 – Present

Saxion University of Applied Sciences — Enschede, Netherlands

- Focus areas: circular textiles, supply chain traceability, Digital Product Passports (DPP), bio-based materials, and life-cycle thinking.
- Selected projects: DPP4CD (with MUD Jeans & Tejidos Royo); Kantamanto Social Club (circular remanufactured textiles); Living Technology (bio-based coatings from food waste); Noosi (responsive textile for dementia care).

BSc Textile Engineering

May 2014 – Dec 2018

BGMEA University of Fashion and Technology (BUFT) — Dhaka, Bangladesh

- Final grade 3.66 / 4.00 (German equivalent: 1.51 – sehr gut).
- Thesis: A comparative study between cotton fibre and kans grass fibre (alternative cellulosic fibre for sustainable textiles).

KEY ACADEMIC PROJECTS

Digital Product Passport for Circular Denim (DPP4CD)

Q3 2025 – 2026

Saxion University — industry collaboration with MUD Jeans (NL) & Tejidos Royo (Valencia, ES)

- Co-developed a Digital Product Passport framework for a circular denim supply chain, spanning yarn, fabric, and dyeing & finishing tiers.
- Built a PEFCR v3.1-aligned yarn-mapping diagram (swim-lane format) and blank data-collection matrices used to gather Scope 3 / LCI data from upstream Tier 2/3 suppliers.
- Authored three A4 Standard Operating Procedures (Yarn; Fabric; Dyeing & Finishing) defining data flows for DPP attributes and traceability hand-offs between supplier tiers.
- Delivered discussion, conclusion, and recommendations slides to industry partners and coordinated written communication with collaborator Tejidos Royo.

Kantamanto Social Club — Circular Supply Chain for Remanufactured Textiles

Q2 2025

Saxion University

- Designed a circular supply-chain concept for remanufactured garments using post-consumer waste flows from the Kantamanto secondhand market (Accra, Ghana), applying multi-tier traceability principles and social-impact considerations.

Living Technology — Repurposing Saxion's Waste Stream (SDG 12)

Current

Saxion University — 4-person team project

- Converting Saxion's institutional waste streams into textile products; individual subtopic: bio-based textile coatings derived from food waste (antimicrobial, water-repellent, flame-retardant, UV-protective and self-cleaning functions).

PROFESSIONAL EXPERIENCE

Senior Sourcing Merchandiser

Dec 2023 – May 2025

GOODEARTH APPARELS LTD. — Gazipur, Bangladesh

- Managed multi-tier supplier relationships across raw-material, fabric, trim, and finishing vendors to ensure consistent quality and on-time delivery for European retail buyers.
- Negotiated pricing, lead times, and commercial terms with vendors to meet cost, delivery, and compliance targets.
- Tracked production from order placement through final delivery, verifying adherence to buyer technical packs and quality standards.
- Collaborated cross-functionally with design & development, production, and logistics teams to translate buyer requirements into deliverable, compliant orders.
- Worked within buyer compliance frameworks covering social and environmental standards (code-of-conduct audits, restricted-substance lists) typical of EU retail customers, and supported supplier evaluation against these requirements.

Executive Merchandiser

Dec 2021 – Nov 2023

GREEN SMART SHIRTS LTD. — Gazipur, Bangladesh

- Primary point of contact with international buyers throughout the order life-cycle; developed products to tech-pack specification.
- Built and maintained Time & Action calendars aligning upstream sourcing with downstream production milestones; followed up raw-material sourcing and procurement.
- Coordinated sourcing of preferred and sustainable material options (e.g., organic cotton, recycled-content fabrics) in line with buyer sustainability briefs.

Assistant Merchandiser

Sep 2019 – Dec 2021

FAKIR APPARELS LTD. — Narayanganj, Bangladesh

- Drove sample development against buyer tech packs; entered and reconciled order data in SAP.
- Coordinated with vendors to in-house raw materials on schedule and to specification.

Intern – Garment Production

Mar 2019 – Sep 2019

NORTHERN TOSRIFA GROUP — Tongi, Bangladesh

- Hands-on training in garment pattern-making and CAD, quality management, and sampling processes.

PUBLICATIONS & CONFERENCES

- Hemal, M.T.I. et al. (2025). Prediction of Standard Minute Value Using Machine Learning in the Garment Industry. Proceedings of the 14th International Conference on Software and Computer Applications (ICSCA 2025), Kuala Lumpur, Malaysia. icsca.org/icsca2025

TECHNICAL & PROFESSIONAL SKILLS

Sustainability & traceability: Life Cycle Assessment (LCA), Digital Product Passport (DPP), GS1/EPCIS, ESPR readiness, CSDDD, Scope 3 / GHG accounting, carbon-footprint analysis, environmental impact assessment, PEFCR, circular economy & Cradle to Cradle principles, multi-tier supply-chain mapping, sustainability reporting, ESG, recycled & organic materials sourcing, stakeholder engagement.

Textile & garment industry: Tech-pack development, multi-tier supplier sourcing & negotiation, sample development, SAP order management, quality management systems, buyer communication.

Software & data: Python (pandas, scikit-learn), Microsoft Office (Excel, Word, PowerPoint), Adobe Illustrator, Adobe Photoshop, CLO 3D, LECTRA Modaris, basic machine-learning workflows.

Soft skills: Cross-cultural stakeholder communication, technical and academic reporting, independent research, multi-project coordination.

LANGUAGES

- **English** — IELTS C1 — fluent in written and verbal communication; proficient across listening, reading, speaking and writing. Working language for all academic and professional output.
- **Bengali** — Native speaker.

ADDITIONAL TRAINING

- Implementing Sustainability Through 3R & Innovation System Approach — Textile Today, Dhaka (Oct 2020).
- Python for Non-Programmers — Multimedia University, Cyberjaya, Malaysia (Apr 2024).